Experiment No-7 : Breadth-First Search (BFS)

Aim

To write a Python program to implement the Breadth-First Search (BFS) algorithm in Artificial Intelligence.

Programming Code:

```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])

    while queue:
        node = queue.popleft()

        if node not in visited:
            print(node, end=" ")
            visited.add(node)

            for neighbour in graph[node]:
                if neighbour not in visited:
                    queue.append(neighbour)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
```

'C': ['F'],

'D': [],

'E': [],

'F': []

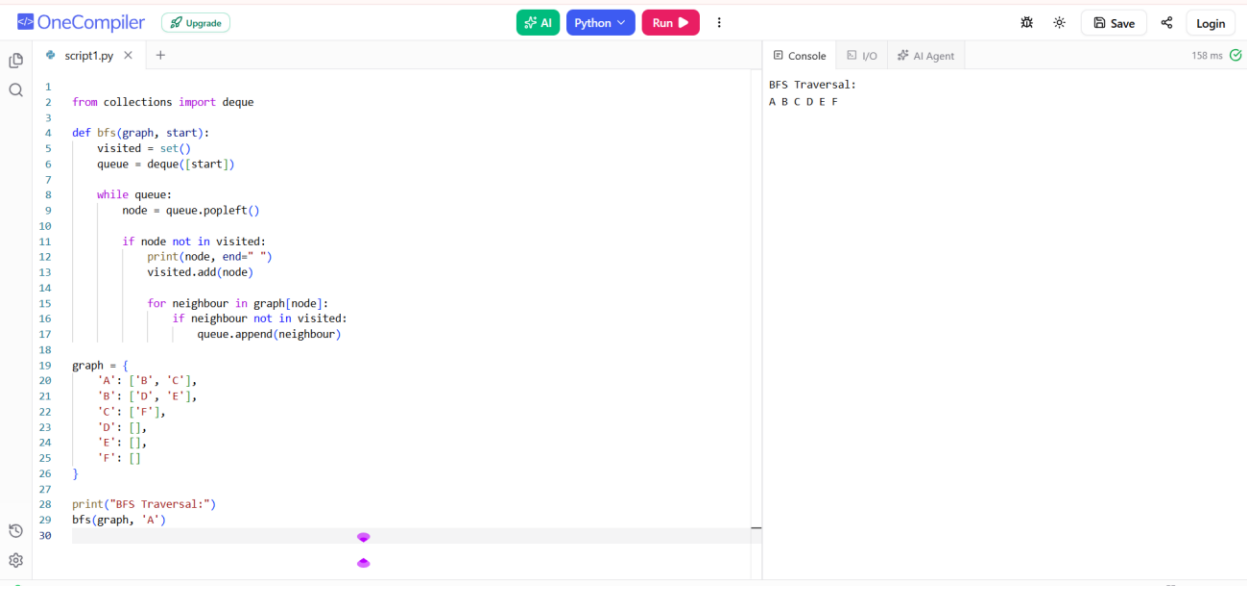
}

print("BFS Traversal:")

bfs(graph, 'A')

OUTPUT:

```
BFS Traversal:
A B C D E F
```



Result

Thus, the Breadth-First Search (BFS) algorithm was implemented successfully in Python. It traverses the graph level by level using a queue.